

Lab 01 - Basic OT-ICS Device Detection



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Lattes

ORCID

SciProfiles

Scopus

ResearcherID

substationworm

LFFreitasGutierrez

Tasks

Warning:

When using specialized search engines or Google dorks to identify Internet-exposed OT devices, **do not interact with or attempt to access any real systems**. The tasks in this document are strictly educational, observational, and non-intrusive, and must fully comply with ethical and legal standards.

- **1** Verify the IP address of the `otlab-student` workstation.
- **2** Determine the subnet range of the network where the `otlab-student` workstation is deployed.
- **3** Discover the IP address, MAC address, and vendor information of other active hosts within the network.

– *Hint: It is a PLC.*

- **4** Identify open ports and available services on the OT-ICS host over both TCP and UDP protocols.
- **5** Determine the proprietary industrial communication protocol used by the PLC.
- **6** Retrieve additional system information via the SNMP protocol.
- **7** Identify the total number of publicly accessible OT-ICS devices using the same proprietary industrial protocol as `conpot-plc` through a specialized search engine such as [Shodan](#) or [FOFA](#).
- **8** Locate a publicly exposed OT device using the same industrial protocol as `conpot-plc` through Google dorking.

Note: The `conpot-plc` is based on [Conpot](#), which remaps standard protocol and service ports to non-privileged ports. Refer to the [link](#) for a list of some default and remapped ports.

Tools

- The following tools are available on the `otlab-student` workstation for completing OT-Lab 1: `ifconfig`, `masscan`, `netdiscover`, `nmap`, and `snmpwalk`.

Nomenclature

- ICS: Industrial control system.
- IP: Internet protocol.
- MAC: Media access control.
- OT: Operational technology.
- PLC: Programmable logic controller.
- SNMP: Simple network management protocol.
- TCP: Transmission control protocol.
- UDP: User datagram protocol.